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Enhancing Critical Thinking: Exploring Human-AI Synergy in Student Cognitive Development

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Abstract
Artificial intelligence (AI) has emerged as a transformative tool, integrated across various sectors. In education, AI has generated significant excitement among students for its potential to enhance learning experiences. However, concerns about overreliance on AI temper this enthusiasm, as it may undermine the development of critical thinking skills. Numerous studies have highlighted the risks associated with students' excessive use of generative AI (GAI) in academic tasks, noting its potential to diminish cognitive abilities. However, the optimal use of AI to enhance students' critical thinking skills remains under-researched. Therefore, this study seeks to answer the question: How does generative AI influence students' critical thinking, self-efficacy, and decision-making? This study aims to explore the synergic relationship between human intelligence and artificial intelligence in augmenting essential thinking skills among students and building upon their existing cognitive resources through self-efficacy, learning motivation, and decision-making. Specifically, it explores the cause-and-effect connections among GAI, self-efficacy, decision-making, learning motivation, and critical thinking skills. A quantitative methodology used an online questionnaire to collect responses from 165 undergraduate, master's, and doctoral students. Statistical analyses, including bootstrapping techniques, were conducted to examine direct and indirect effects. The results revealed that GAI has a significant positive influence on self-efficacy, learning motivation, decision-making, and critical thinking skills. In turn, self-efficacy, learning motivation, and decision-making significantly impact critical thinking skills. The mediating results indicated that GAI can indirectly boost students' critical thinking by enhancing self-efficacy, learning motivation, and decision-making. This suggests that AI capabilities can transform the cognitive learning process.

Keywords: Critical thinking skills, decision making, generative artificial intelligence, learning motivation, plagiarism, self-efficacy, student cognitive development

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Introduction

The sudden shift to online learning after the COVID-19 pandemic has been challenging for many university students. This disruption, coupled with fewer opportunities for social interaction, has negatively affected students' cognitive and emotional well-being, potentially hindering the development of critical thinking skills and deep engagement with learning materials. Learning has been digitized post-pandemic to ensure the continuity of learning during lockdown. This has led to the launch of artificial intelligence, which has profoundly influenced various aspects of pedagogical practices in higher education. Generative AI, exemplified by technologies like ChatGPT, has firmly established itself within educational settings. This prominence is attributed to its diverse capabilities encompassing tasks such as generating text-based responses, assisting in evaluating and analyzing data, completing essays and academic projects, providing timely feedback, and organizing thoughts more effectively. ChatGPT aids students in brainstorming ideas and strategies and provides them with effective resources for their academic work. (Dai et al., 2023; Berg, 2023) Furthermore, generative AI can provide students with examples and alternative perspectives. However, numerous studies have raised serious concerns regarding the overdependence on AI tools such as ChatGPT. Many previous researchers have highlighted the potential harm of AI tools in diminishing students' critical thinking abilities. These tools can be tempting shortcuts, leading some students to rely on AI-generated content rather than developing their critical analysis. Plagiarism is another negative aspect of AI tools, as they can induce students to engage in unethical academic practices. This could undermine the trust and credibility of higher education institutions. However, the synergy of human intelligence and generative AI can boost critical thinking abilities when students critically evaluate and build on AI-generated information. It is quite fundamental for learners to initially develop their hypothesis, then validate their assumptions through using generative AI. However, it's crucial to remain vigilant in guiding the AI to improve its assistance and utilizing its feedback and guidance to enhance and optimize your work effectively.

Therefore, this study aims to investigate the synergistic impact of integrating human intelligence with generative AI on enhancing critical thinking abilities among students. The research aims to explore how generative AI augments human critical thinking and complements students' existing cognitive resources through self-efficacy, learning motivation, and decision-making. In other words, this study seeks to examine how self-efficacy, learning motivation, and decision-making mediate the relationship between the use of AI tools and the development of critical thinking skills in educational settings. The following questions guided this study:

1. How does the collaboration between human intelligence and artificial intelligence enhance students' critical thinking skills?
2. To what extent do AI tools influence students' self-efficacy, learning motivation, and decision-making in academic contexts?
3. How do students perceive the impact of AI tools on their critical thinking skills when used alongside their cognitive abilities?

Literature Review

Generative AI and Education

Generative AI tools such as ChatGPT are language models that are designed to generate human-like content based on the instructions they receive. During its nascent stage, it was equipped to mimic human language by emulating human syntax, semantics, and grammar. It can also engage in conversations with users and generate output for inquiries. ChatGPT is designed to allow users to initiate and control the direction of the dialogue.

Shanto and Ahmed (2024) posit that to balance generative AI and human cognition, students must actively manage their inquiries by crafting effective prompts, seeking diverse perspectives from AI tools, assessing responses, synthesizing insights, and engaging in reflective learning. In essence, instead of passively consuming information, students are encouraged to interact critically with AI-generated content, exercising their cognitive agency by fact-checking and participating in a dynamic negotiation process. This mutual observation and adjustment between humans and AI ensures a collaborative and adaptive interaction, and most importantly, promotes critical thinking. However, Dai et al (2023) argued that students need to possess a range of skills for conducting and overseeing inquiries because ChatGPT output is significantly shaped by the inputs provided by students.

Several previous studies have highlighted the importance of generative AI tools and their role in contributing to education in general. Van den Berg and Elize (2023) emphasize the potential of generative AI-powered tools to assist educators. These tools can offer support by generating lesson plans, brainstorming objectives and strategies, and facilitating the collection of pertinent teaching materials. As a result, this can enhance teachers' critical thinking skills, compelling them to explore diverse approaches to their lessons. However, their research emphasizes the importance of caution when employing these models by evaluating their limitations and potential biases (Meniado, 2023). In a similar vein, Javaid et al. (2023) argued that students can benefit from ChatGPT, as it provides them with timely feedback and helps them identify logical inconsistencies and fallacies in their reasoning, which in turn boosts their critical thinking. Berg (2023) has proposed three promising applications for AI tools in education. These encompass AI's role as a mentor, its capacity for analyzing academic content, and its ability to enhance the efficiency of academic writing processes.

However, the extensive use of generative AI tools among students can have negative consequences. Abbas et al. (2024) argue that overreliance on tools like ChatGPT can harm students' learning abilities, leading to memory loss, procrastination, and reduced academic performance. However, when used responsibly, ChatGPT and similar AI tools can offer significant benefits. Dai et al. (2023) suggest that when students actively engage with AI tools, it can foster motivation and enhance critical thinking by facilitating brainstorming and generating diverse perspectives. Essel et al. (2022) found that GAI boosted students' self-efficacy and learning outcomes, while Lee et al. (2022) noted improvements in academic performance, self-efficacy, attitude, and motivation. Additionally, Essien et al. (2024) and Atlas (2023) emphasized that GAI, like ChatGPT, can promote the development of critical thinking skills by offering alternative scenarios and insights. Chuma and Oliveira (2023) reported that students agreed GAI significantly enhanced decision-making by expanding their initial insights (Chan & Hu, 2023).

Conceptual Framework and Research Hypotheses

Critical Thinking Skills

Critical thinking is essential in higher education, shaping students' learning, analysis, reasoning, and decision-making (Hwang et al., 2018). To succeed academically and enhance employability, students need higher-order thinking skills for problem-solving and creativity (Krathwohl, 2002). Critical thinking enables students to evaluate, analyze, and question information rather than passively consuming it. Many educators stress the need to integrate these skills into curricula. However, the rise of AI, particularly Generative AI (GAI), has raised concerns about its impact on critical thinking, as students may over-rely on it (Cooper, 2023; Chan & Hu, 2023; Ding et al., 2023). Nevertheless, if human intelligence and GAI were unified, they could potentially enhance critical thinking abilities rather than impede their development. Research highlights that if GAI capabilities are used effectively and collaborate with human intelligence, they can enhance critical thinking and complement human cognition (Spector, 2019). Generative AI can be a valuable tool in education as it facilitates the creation of intricate scenarios that engage students in higher-order thinking processes. Put simply, the impact of Generative AI (GAI) on critical thinking revolves around learners leveraging the diverse perspectives generated by the AI to enhance their own understanding and analytical abilities. Numerous prior studies have concentrated on the potential negative effects of Generative AI (GAI) on learners' higher-order thinking skills. Yet, little attention has been paid to its influence on the development of critical thinking abilities. Thus, this study endeavors to explore whether AI contributes positively to the enhancement of critical thinking skills among college students. Consequently, the following hypothesis is developed:

H1: GAI has a significant impact on critical thinking abilities.

Self-efficacy

Many previous research underscores that GAI has a significant impact on learners' self-efficacy. According to Bandura (1977), self-efficacy is the belief individuals have about their capabilities to perform specific actions or tasks. Furthermore, research indicates that self-efficacy helps reduce the impact of stressors on perceived stress levels among college students. It also emerges as a predictor of academic success and contributes to a decrease in procrastination tendencies. (Zajacova et al, 2005; Ren et al, 2021) GAI systems can provide immediate feedback on users' decisions and actions, allowing them to see the direct impact of their choices and efforts. Positive feedback from successfully using GAI tools can reinforce individuals' beliefs in their competence and effectiveness, thereby boosting self-efficacy. (Yilmaz, 2023; Jia & Tu 2024; Wang et al., 2023). Also, working collaboratively with GAI systems can foster a sense of partnership and shared responsibility in problem-solving endeavors. As individuals witness the combined impact of their efforts and the capabilities of AI technology, they may develop a stronger belief in their ability to achieve positive outcomes through collaborative action. GAI systems can adapt to users' preferences and behaviors, providing personalized assistance tailored to their individual needs and learning styles (Dai, 2023; Bouzar, 2024). To this end, the present study intends to reveal whether GAI affects students' self-efficacy, hence, this hypothesis is developed:

H2: GAI has a significant influence on self-efficacy.

Learning Motivation

Learning motivation encompasses both the internal and external factors that encourage individuals to be involved with learning activities. When the learning motivation is high, people tend to have the desire, interest, and willingness to learn and acquire new knowledge, skills, or competencies. Many prior studies have emphasized the impact of learning motivation on academic performance. In the realm of education and learning motivation, Researchers consider GAI to have high potential for motivating students to learn, because GAI is designed to provide immediate feedback on students' performance, allowing them to monitor their progress and adjust their learning strategies accordingly. (Yin et al.,2020; Kamita et al., 2019; Popenici & Kerr 2017).

Timely feedback helps students stay motivated and focused on their learning goals. Jia and Tu (2024) found in their study that the integration of AI has significantly influenced students' motivation to learn. This impact is evident not only in providing personalized learning experiences but also in areas such as instant feedback and the ability to learn at one's own pace. Yin et al. (2020) conducted a quasi-experiment on how AI chatbots boost learning motivation. They concluded that AI chatbot learning systems effectively increase students' intrinsic learning motivation compared to conventional learning.
H3: GAI has a significant impact on learning motivation.

Decision Making

Decision-making is a key component of higher-order thinking skills. When learners engage in tasks that require critical thinking, they need to question and evaluate the validity of AI-generated content, ensuring the accuracy of information to make informed decisions autonomously. (Spector, 2019) In essence, GAI can act as a mentor who can tutor and provide learners with access to a wide range of information and perspectives relevant to their decision-making process. (Berg, 2023). In other terms, GAI provides guidance and suggestions to assist learners in their decision-making process without imposing predetermined solutions by offering options, recommendations, and thought-provoking insights. Engaging students in brainstorming sessions with Generative AI (e.g., ChatGPT) can spark creative ideation (Doering, 2015). Inquiry-based learning promotes curiosity and the exploration of diverse perspectives. However, it's imperative to acknowledge the inherent limitations of GAI models, which may inadvertently introduce biases or disseminate misleading information due to their lack of human judgment. It is the responsibility of learners to critically discern potential biases in AI-generated content and refrain from incorporating skewed information into their work or decision-making processes. Proficiency in navigating inquiry with GAI learning tools empowers students to pose insightful questions, explore various viewpoints, and formulate their own hypotheses, thereby sharpening their critical thinking skills. Consequently, students are encouraged to actively gather a broad spectrum of ideas and rigorously analyze the underlying rationale behind them. Additionally, GAI encourages learners to consider the strengths and weaknesses of their decisions, fostering metacognitive awareness and critical self-reflection. (Essien 2024). In summary, these previous studies emphasized the affirmative contribution of AI in augmenting student decision-making abilities. This leads to the following hypothesis.
H4: GAI has a positive impact on learners' decision-making.

The Impact of Self-efficacy, Learning motivation, and Decision-making on Critical thinking

Many previous studies have focused on the relationship between self-efficacy and critical thinking, demonstrating that those with stronger self-efficacy in their academic ability display higher levels of critical thinking skills. (Jia and Tu, 2024)

In other words, students who are confident in their abilities to excel in their academic performance tend to evaluate information and raise thought-provoking questions (Bandura, 1995; Dehghani, 2011; Jia & Hu, 2024; Vachova et al., 2021). Also, previous studies have highlighted the correlation between self-efficacy and learning motivation (Zimmerman, 2000; Hasanah et al., 2019). Individuals with high self-efficacy believe in their ability to perform tasks successfully. This sense of competence fuels their motivation because they feel confident in achieving their goals.

Learning motivation influences critical thinking skills significantly. Students with high academic motivation are more eager to engage in problem-solving and critical-thinking activities that sharpen their cognitive abilities. This engagement not only boosts their critical thinking abilities but also strengthens their academic performance. This proactive approach fosters a conducive environment for developing critical thinking skills. Studies have found that decision-making is one of the main components of critical thinking as it requires students to analyze and assess information impartially, resulting in well-informed decisions. Decision-making requires students to systematically gather and evaluate information, weigh the pros and cons of different options, and consider potential outcomes. This process is inherently analytical and reflective, which are key aspects of critical thinking. By practicing effective decision-making, students enhance their ability to think critically, as they learn to approach problems and consider various perspectives. (Ennis, 1993; Brookhart, 2010; Turan et al.,2019; Ding et al., 2023). Overall, the previous studies support a positive impact of general self-efficacy, learning motivation, and decision-making on critical thinking ability. Therefore, this study develops the following hypotheses:
H5: Self-efficacy has a positive impact on critical thinking skills
H6. Self-efficacy has a positive impact on learning motivation
H7: Learning motivation has a positive impact on critical thinking abilities
H8: Decision-making has a positive impact on critical thinking abilities

The Mediating role of Self-efficacy, Learning motivation, and Decision-making

This study further proposes that self-efficacy, learning motivation, and decision-making will mediate the relationships between GAI use and critical thinking. As mentioned above, learners collaborating with GAI to complete their academic tasks are more likely to develop self-efficacy, learning motivation, and decision-making abilities. In turn, these factors contribute to developing critical thinking abilities as students learn to analyze, synthesize, and evaluate information systematically and reflectively. Hence, the following hypotheses are developed:
H9. Self-efficacy will mediate the relationship between the use of GAI and critical thinking skills.
H10. Learning motivation will mediate the relationship between the use of GAI and critical thinking.
H11. Self-efficacy and learning motivation will mediate the relationship between GAI and critical thinking.
H12. Decision-making will mediate the relationship between the use of GAI and critical thinking.

Method

This study aims to investigate how generative AI, along with human intelligence, augments students' cognitive and critical thinking skills. Therefore, this study aims to assess whether factors such as self-efficacy, learning motivation, and decision-making are boosted through the use of generative AI in learning and ultimately lead to stronger critical thinking skills. The research methodology for this study involves six sequential steps: conducting a literature review, developing research hypotheses, collecting data, designing and validating questionnaires, performing statistical analysis, and discussing the results. This study used a quantitative approach to analyze the research findings objectively and ensure reliability.

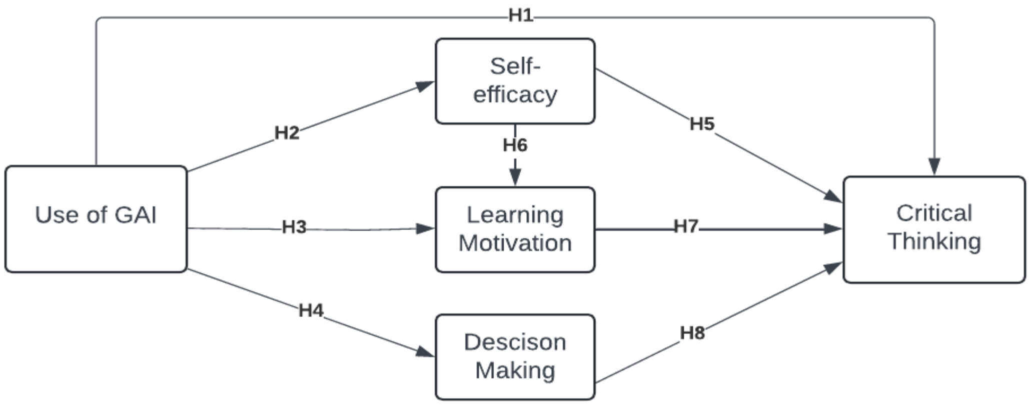

Figure 1. The proposed research model

Participants and Procedures

The study employed random sampling techniques, involving 165 students from undergraduate, master's, and doctoral programs in the English department at the Faculty of Letters and Human Sciences of a university in Morocco. Data were gathered through an online questionnaire hosted on Microsoft Forms, ensuring security via Office 365. Distribution utilized popular platforms such as WhatsApp and Gmail to maximize participation. Participants were assured of confidentiality. Statistical analysis was conducted using SPSS version 25.0, ensuring thorough examination and interpretation of data to uphold the study's reliability and validity.

Instruments

The research instrument employed in this study was a structured questionnaire designed to align with the study's objectives and existing literature. The questionnaire comprised five sections: demographic information, critical thinking, self-efficacy, learning motivation, and decision-making. The demographic section collected respondents' sex, age, education level, familiarity with generative AI, and frequency of generative AI usage, using multiple-choice questions. The critical thinking section included four items, the self-efficacy section had three items, the learning motivation section comprised three items, and the decision-making section contained four items. All sections utilized a Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree), with 14 items across the questionnaire. See Table 2.

Results

Demographic and Usage Characteristics of Participants

Table one presents the participants' demographic characteristics and usage patterns of generative AI technologies.

Table 1. *The Profile of Participants*

Item	Category	Frequency	Percentage (%)
Gender	Male	72	43,6
	Female	93	56,4
Age	18-24	76	46,1
	25-30	65	39,4
	Above 30	24	14,5

Educational background	Undergraduate	34	20,6
	Master	57	34,5
	Doctoral	74	44,8
Have you ever used generative AI technologies like ChatGPT?	Never	3	1,8
	Occasionally	23	13,9
	Sometimes	94	57,0
	Often	29	17,6
	Always	16	9,7
Total		165	100

This study examined Participants' demographic attributes and usage patterns of generative AI technologies to understand their profile and engagement with AI-driven tools. The sample comprised 165 individuals, predominantly female (56.4%) compared to male participants (43.6%). Age distribution indicated that a substantial proportion fell within the 18-24 years category (46.1%) and 25-30 years category (39.4%), with a smaller segment aged above 30 years (14.5%). Educational backgrounds varied, with doctoral students representing the largest group (44.8%), followed by master's students (34.5%) and undergraduate students (20.6%). Regarding the use of generative AI technologies such as ChatGPT, a majority of participants reported using them sometimes (57.0%), followed by often (17.6%), occasionally (13.9%), always (9.7%), and very few never used them (1.8%)

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Measurement Model Assessment

Factor analysis was conducted to justify the scales of different variables used in this study. The internal consistency reliability was tested using composite reliability (CR) and Cronbach’s alpha. The analysis revealed that the composite reliability, scores of all the constructs were above the minimum threshold (CR> 0.70). Also, Cronbach’s alpha of each item exceeded 0.7. Hence, indicating construct reliability.

Table 2. *Reliability and Convergent Validity of the Measurement Model*

Items	Loadings	CA	CR	AVE
Critical Thinking		0,904	0.828	0.548
1. Generative AI tools like ChatGPT encourage me to think creatively and explore different perspectives while learning	0,663			
2. AI-generated information helps me identify the strengths and weaknesses of my reasoning	0,737			
3. Generative AI helps me organize my thoughts more effectively	0,804			
4. Using generative AI tools has encouraged me to evaluate sources and evidence critically.	0,751			
Decision Making		0,873	0.830	0.549
1. Generative AI (ChatGPT) supports me in making informed decisions related to my learning.	0,737			
2. Generative AI tools like ChatGPT help me brainstorm a broader range of possible solutions or options.	0,750			
3. I feel more confident in forming my conclusions after using generative AI.	0,717			
4. Generative AI tools like ChatGPT help me explore different possibilities and approaches when deciding.	0,761			
Self Efficacy		0,914	0.799	0.574

1. Using generative AI tools enhances my confidence in solving problems independently.	0,686			
2. Generative AI tools help me overcome challenges in understanding complex topics.	0,693			
3. I am confident that using generative AI (ChatGPT) enhances my learning efficiency	0,878			
Learning Motivation		0,844	0.794	0.564
1. Generative AI tools make learning or problem-solving activities more interesting and engaging.	0,752			
2. To what extent does collaborating with AI in learning motivate you to engage in educational activities?	0,669			
3. Using generative AI tools motivates me to ask more questions and seek additional information to evaluate the information presented	0,824			

$p < 0.05$, CA Cronbach's Alpha, CR composite reliability, AVE average variance extracted.

A critical step was evaluating the overall significance of the correlation matrix using Barlett's test of sphericity, which assesses whether there are significant correlations among its components. The results were significant, $\chi 2$ (n=165) = 1985,420 ($p < 0.01$), which indicated its suitability for factor analysis. The Kaiser-Meyer-Olkin measure of sampling adequacy was 0,944. In this regard, data MSA values above 0.8 are considered appropriate for factor analysis. Additional examinations revealed that the Average Variance Extracted (AVE) for the five constructs meets the recommended minimum threshold of 0.50, suggesting a satisfactory level of convergent validity (Fornell & Larcker, 1981; Hair, 2010).

Correlation Analysis

Pearson's correlation analysis was performed to measure the direction and strength between variables measuring the use of generative AI, critical thinking, decision-making, self-efficacy, and learning motivation among the participants. The results are summarized in Table 3. Table 3 presents a comprehensive overview of the correlation analysis, revealing the interplay among key variables. Notably, all correlation coefficients demonstrate positive associations. First, the use of generative AI was found to be positively correlated with critical thinking ($r = 0.241$, $p < 0.05$), self-efficacy ($r = 0.247$, $p < 0.05$), and learning motivation ($r = 0.260$, $p < 0.05$). These findings suggest that individuals who engage with generative AI tools perceive enhancements in critical thinking abilities, self-efficacy beliefs, and motivation for learning tasks.

Table 3. *Correlation Analysis*

		1	2	3	4	5
1	Use of generative AI	1				
2	Critical thinking	,241** 002	1			
3	Decision making	,199** 010	,841** 000	1		
4	Self-efficacy	,247** 001	,786** 000	,823** 000	1	
5	learning motivation	,260** 001	,683** 000	,730** 000	,748** 000	1
M		3,19	3,93	3,89	3,97	3,75
SD		0,86	0,86	0,81	0,86	0,85
Skewness		,366	-,943	-,838	-1,359	-,762

Kurtosis		,372	1,25	,945	2,362	,817

**Correlation is significant at the 0.01 level (2-tailed). M: Mean, SD: Standard Deviation.

However, the correlation between generative AI and decision-making is weak ($r = 0.199$, $p < 0.05$). Indicating that while individuals who engage with generative AI tools tend to report higher levels of decision-making ability, the extent of this relationship is relatively mild. This implies that while the correlation between the use of generative AI and decision-making is statistically significant, the strength of this relationship is modest. Moreover, critical thinking demonstrated strong positive correlations with decision-making ($r = 0.841$, $p < 0.01$), self-efficacy ($r = 0.786$, $p < 0.01$), and learning motivation ($r = 0.683$, $p < 0.01$), highlighting its integral role in fostering adaptive cognitive and motivational outcomes. Decision-making, in turn, was significantly associated with self-efficacy ($r = 0.823$, $p < 0.01$) and learning motivation ($r = 0.730$, $p < 0.01$), indicating its influence on both personal efficacy perceptions and motivation to engage in educational activities. Furthermore, self-efficacy showed a robust positive correlation with learning motivation ($r = 0.748$, $p < 0.01$), underscoring the importance of confidence in one's abilities in fostering a proactive approach to learning tasks. These findings contribute to understanding how the integration of generative AI technologies may positively impact cognitive processes and motivational dynamics in educational settings. The mean (M), standard deviation (SD), skewness, and kurtosis of the variables were measured to provide insights into how students perceive the influence of generative AI on critical thinking, self-efficacy, decision-making, and learning motivation. Participants reported a moderate mean score of 3.19 ($SD = 0.862$) for the use of generative AI, indicating a generally positive but not highly frequent engagement with AI-driven tools.

Critical thinking received the highest mean score ($M = 3.9348$, $SD = 0.86730$), suggesting an essential level of critical thinking reported among the students who use generative AI. Decision-making ($M = 3.8985$, $SD = 0.81614$), self-efficacy ($M = 3.9717$, $SD = 0.86673$), and learning motivation ($M = 3.7596$, $SD = 0.85269$) also exhibited moderate mean scores, reflecting balanced levels of perceived decision-making ability, self-confidence, and motivation for learning tasks.

Skewness and kurtosis statistics provide additional insights into the distributional characteristics of the variables. Skewness values close to zero (use of generative AI, critical thinking, decision making) indicate approximately symmetric distributions. In contrast, negative skewness in self-efficacy (-1.359) and learning motivation (-0.762) suggests slight left-skewness, indicating some students may report higher scores in these constructs.

Hypothesis Testing

Using bootstrapping procedures, we tested the study's hypotheses for both direct and indirect effects. The number of bootstrap subsamples was set to 5000 to ensure the robustness and reliability of the results. As presented in Table **four,** the findings indicated a significant positive relationship between the use of GAI and critical thinking ($\beta = 0.243$, $t = 3.175$, $p = 0.002$), suggesting that students who engage with GAI tools are likely to experience enhanced critical thinking abilities. Thereby supporting hypothesis 1. Similarly, using GAI was found to significantly predict self-efficacy ($\beta = 0.248$, $t = 3.252$, $p = 0.001$). This result implies that the use of GAI tools contributes positively to students' confidence in their abilities to perform tasks and solve problems. Thereby, hypothesis 2 is accepted. Furthermore, the hypothesis testing for the impact of GAI on learning motivation revealed a significant positive relationship ($\beta = 0.257$, $t = 3.441$, $p = 0.001$). This emphasizes that GAI use is associated with increased motivation to engage in learning activities. Consequently, hypothesis 3 is accepted. Hypothesis 4 tested whether the use of generative AI (GAI) significantly influences students' decision-making abilities. The analysis indicated a marginally significant positive relationship between GAI and decision-making (β = 0.189, t = 2.598, p = 0.01). This result suggests that students who utilize GAI tools tend to have improved decision-making skills. Therefore, H4 is supported.

Table 4. *The Results of Hypothesis Testing*

Hypothesis	Path	Coefficient	T-value	F	p-value	Status
H1	Use of GAI -> Critical thinking	0,243	3,175	10,083	0,002	Supported
H2	Use of GAI -> Self-efficacy	0,248	3,252	10,577	0,001	Supported
H3	Use of GAI -> Learning motivation	0,257	3,441	11,839	0,001	Supported
H4	Use of GAI -> Decision making	0,189	2,598	6,749	0,01	Supported
H5	Self-efficacy -> Critical thinking	0,787	16,252	264,137	0	Supported
H6	Self-efficacy -> learning motivation	0,736	14,387	206,978	0	Supported
H7	Learning motivation -> Critical thinking	0,695	11,937	142,491	0	Supported
H8	Decision making-> critical thinking	0,894	19,867	394,711	0	Supported

The final set of hypotheses aimed to explore the impact of self-efficacy, learning motivation, and decision-making on critical thinking abilities among students. Hypothesis 5 posited that self-efficacy would positively influence critical thinking. The results strongly supported this hypothesis, revealing a significant positive relationship (β = 0.787, t = 16.252, p < 0.001). The high coefficient value indicates that students with higher self-efficacy are likely to exhibit more potent critical thinking abilities. As a result, hypothesis 5 is supported. Consistent with hypothesis 5, self-efficacy has a significant positive effect on learning motivation (β=0.748, t=14.387t = 14.387 p < 0.001). This suggests that higher self-efficacy is strongly associated with increased learning motivation among students. Thus, hypothesis 6 is accepted.

The finding further examined the effect of learning motivation on critical thinking. The analysis showed a significant positive relationship (β = 0.695, t = 11.937, p < 0.001), suggesting that increased learning motivation is associated with enhanced critical thinking skills. The coefficient value, while slightly lower than that of self-efficacy’s, still indicates a substantial impact. The F-value for this relationship was 142.491, further affirming the strength and significance of the model. Thus, hypothesis 7 is also accepted. Finally, Hypothesis 8 investigated whether decision-making skills predict critical thinking abilities. The findings indicated a robust positive relationship (β = 0.894, t = 19.867, p < 0.001). The coefficient value for this relationship is the highest among all the examined paths, suggesting that decision-making skills are a critical determinant of critical thinking. The F-value for this relationship was 394.711, indicating an exceptionally strong model fit and highlighting the profound impact of decision-making on critical thinking. Hence, hypotheses 7 find strong empirical support in the data.

Table fiv**e** presents the results of all indirect effects. As shown in Table **five**, the study investigated the mediating role of self-efficacy, learning motivation, and decision-making in the associations between the use of generative AI and critical thinking. The results indicated noteworthy indirect effects, specifically, the path from GAI use to self-efficacy, and subsequently to critical thinking, yielded a significant indirect effect (β= 0.192, t = 2.767), supporting the hypothesis that self-efficacy mediates this relationship. This posits that self-efficacy plays a mediating role in enhancing critical thinking abilities when individuals engage with GAI tools.

Similarly, using generative AI has a positive indirect relationship with critical thinking (β = 0.174, t = 2.855) through learning motivation. Additionally, the mediation analysis reveals that both self-efficacy and learning motivation significantly mediate the relationship between generative AI use and critical thinking. Specifically, the indirect effect through self-efficacy is significant (β= 0.154, t= 2.846), suggesting that generative AI enhances self-efficacy, boosting critical thinking abilities. Additionally, the indirect effect through learning motivation is marginally significant (β = 0.054, t=1.785), highlighting that generative AI fosters learning motivation, thereby contributing to improved critical thinking.

Table 5. *Significant Mediating Effects of the Research Model*

Path	Coefficient effect	Boot SE	Confidence Interval	T-value	Status
Use of GAI -> self-efficacy ->critical thinking	0,192	0,0695	,0546,3308	2.767	Supported
Use of GAI-> learning motivation->critical thinking	0,174	0,0607	,0635,3036	2.855	Supported
Use of GAI-> self-efficacy -> learning motivation-> critical thinking	0,154 0.054	0,0541 0,0307	,0508 ,2629 ,0077 ,1261	2.846 1.785	Supported
Use of GAI -> decision making ->critical thinking	0,165	0,071	,0282 ,3063	2.333	Supported

This finding implies that students' increased learning motivation, potentially facilitated by GAI interactions, contributes positively to critical thinking skills. Decision-making is mediating the Use of GAI and critical thinking; the results showed a significant indirect effect (β = 0.165, t = 2.333), supporting the idea that students who engage with generative AI in their learning processes demonstrate improved decision-making abilities. This, in turn, appears to cultivate the development of critical thinking skills. The mediation status is further affirmed because none of the confidence intervals included zero. These findings underscore the significance of self-efficacy, learning motivation, and decision-making in developing critical thinking abilities among students collaborating with GAI to carry out their academic tasks.

Discussion

The primary aim of this research was to explore how generative AI augments human critical thinking and complements students' existing cognitive resources, specifically through the mediating roles of self-efficacy, learning motivation, and decision-making. The findings of this study provide compelling evidence that generative AI can significantly enhance these cognitive resources, thereby promoting more effective critical thinking skills among students.

Despite much previous research found that artificial intelligence minimalizes students' abilities for in-depth analysis and logical reasoning (Abbas et al., 2024; Jia &Tu, 2024; Igbokwe, 2023) on the basis that critical thinking skills abilities require high-order thinking functions, analyzing, evaluating and synthesizing information to make reasoned and well-thought-out decisions or judgments (Muthmainnah, 2022), and generative AI exemplified by ChatGPT undermine students cognitive functioning, and critical thinking abilities (Bahrini et al., 2023; Dwivedi et al., 2023). However, this study demonstrates that critical thinking encompasses a thorough analysis, reasoning, and information critique. It relies not only on human reflection and practical experience but also on AI, which can be effective when GAI capabilities are leveraged. When AI collaborates with human intelligence, it can enhance critical thinking and complement human cognition. The findings of H1 reveal that AI contributes positively to enhancing critical thinking skills among college students, supporting the idea that Generative AI can be a valuable tool in education as it facilitates the creation of intricate scenarios that engage students in higher-order thinking processes. In a similar vein, H2 was supported, validating the primary effect of AI capabilities on students' self-efficacy. (Chen et al., 2023; Li et al., 2021). Thus, partnering with GAI has a favorable mental influence on students, enhancing their general self-efficacy and helping them feel more secure in dealing with academic issues. (Jia &Tu, 2024; Dong et al., 2022). Furthermore, the findings indicated that GAI positively influences learning motivation, supporting H3. This result aligns with the idea that timely feedback keeps students motivated and focused on learning objectives. AI offers tailored pacing and corrective feedback, enabling students to see their progress more quickly. These immediate feedback mechanisms can substantially boost students' motivation to learn. Similarly, H4 found empirical support in the findings, GAI significantly influences students’ decision-making. It is apparent that GAI gives learners access to a broad spectrum of information and perspectives pertinent to their decision-making process (Berg, 2023). Hence, GAI provides guidance and suggestions to assist learners in their decision-making process without imposing predefined solutions, inducing students’ decision-making abilities to make sound judgments. Furthermore, the results indicate a significant correlation between self-efficacy and critical thinking, suggesting that individuals with higher confidence in their academic abilities also demonstrate enhanced critical thinking awareness. Likewise, learning motivation was found to be a significant antecedent of critical thinking, indicating that students who are highly motivated academically tend to show a strong inclination towards engaging in problem-solving and critical thinking processes. This intrinsic motivation often translates into a drive for perfectionism and positively impact cognitive development and academic performance (Jia and Tu, 2024). Thus, the results support that learning motivation has a significant influence on critical thinking skills. Decision-making is found to be the highest determinant that influences critical thinking. These findings indicate that effective decision-making involves cognitive processes such as problem analysis, information evaluation, and logical reasoning, which are core components of critical thinking.

The statistical analysis of the mediating effects suggests that self-efficacy, learning motivation, and decision-making serve as intermediaries in the relationship between the use of artificial intelligence and the development of critical thinking skills. It is important to note that the mediating effects of self-efficacy, learning motivation, and decision-making constitute complete mediation. The results indicated that the use of generative AI positively influences self-efficacy, which in turn enhances critical thinking. This suggests that when students feel more confident in their abilities, thanks to the supportive capabilities of AI tools, they are better equipped to engage in complex critical thinking tasks. The significant indirect effect stresses the importance of self-efficacy as a key mediator in this relationship. Moreover, the study revealed that learning motivation is significantly boosted by the use of generative AI, which subsequently improves critical thinking skills. The observed indirect effect highlights how generative AI not only supports students in their learning but also motivates them to engage more deeply with the material, leading to deeper and more critical engagement with the material. The findings also underscore the importance of self-efficacy and learning motivation as key mechanisms through which generative

AI influences critical thinking, suggesting that when students utilize generative AI for learning, they demonstrate increased self-efficacy and confidence in solving complex tasks (Jia & Tu, 2024). This increased confidence directly boosts their drive to succeed academically, as they believe in the positive outcomes of their efforts. Such a proactive mindset can aid in cultivating and improving critical thinking abilities. This proactive approach enables the development of critical thinking skills, as students become keener to analyze and critique ideas. Additionally, the research demonstrated a significant indirect effect of generative AI on critical thinking through enhanced decision-making abilities. This finding suggests that integrating AI in educational settings helps students develop better decision-making skills, which are crucial for critical thinking. By providing students with tools to analyze and evaluate information more effectively, generative AI contributes to their overall cognitive development.

Conclusion

The main aim of this study is to investigate the synergic relationship between GAI and human intelligence to promote critical thinking skills. The results revealed that GAI indirectly promotes critical thinking awareness by boosting self-efficacy, learning motivation, and decision-making.

While recognizing the potential risks AI may pose to students' cognitive abilities, it is crucial to highlight the benefits when students actively engage with AI-generated content. Students can enhance their academic performance through questioning, analyzing, and critiquing AI-generated output. This active engagement cultivates increased self-efficacy, a positive learning attitude, motivation, and improved decision-making skills. Together, these factors contribute significantly to the development of higher-order thinking skills. These results underscore the potential of GAI, when effectively integrated into learning, to positively influence students' cognitive development and critical thinking.

Limitations and Future Research

Although the findings of this study confirm that the students' integration of GAI in their learning directly enhances their critical thinking, this result is marginally significant. Therefore, due to the uncertainty in the strength of the effect of using GAI on critical thinking skills, it is advised to exercise caution when interpreting the results, and they should not be seen as conclusive. Future research should replicate and validate the study to enhance the understanding of how the variables in the proposed path model interact and mutually influence one another. Moreover, this study targeted participants from one university in Morocco. Therefore, caution should be exercised when generalizing these findings to other institutions. Future research could strengthen the validity of the results by including participants from various educational settings.

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